

Worksheet 4 — The Price of Looking

STAT 481 — Multivariate Analysis • Lecture 4 • Tue, Sep 1, 2026

1. Objective of this lecture

By the end of this sheet you should be able to:

1. turn “how many variables can I look at?” into an arithmetic question with an exact answer, given a stated budget;
2. say how far a single correlation wanders when the two columns have nothing to do with each other, *predict it from n before measuring it*, and then predict how far the *largest* of m such correlations wanders;
3. use that number as a floor, apply it to a real 175×52 panel, and say what is left of the panel’s “findings” once the floor and the definitions are taken out.

2. The data, and how to get it in R

Data set: `whwide2022.csv`, in the course project — no package needed, `read.csv()` is base R. **Size:** the *same* 175 countries as `worldhealth2022`, in the same row order, on **52** World Bank indicators for 2022. Columns 1–4 are `iso3`, `country`, `region`, `income`; columns 5–56 are the indicators, and the first eight of those are the spine you already know.

```
w <- read.csv("data/whwide2022.csv", stringsAsFactors = FALSE)
X <- w[, 5:56] # the 52 indicators
n <- nrow(X); p <- ncol(X)
R <- cor(X, use = "pairwise.complete.obs")
rv <- abs(R[upper.tri(R)]) # the 1326 distinct |r|
c(n = n, p = p, pairs = length(rv))
```

n	p	pairs
175	52	1326

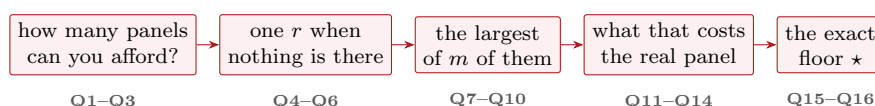
family	example columns	#	family	example columns	#
health	life_exp, birth_rate	6	communications	mobile, broadband	4
population	pop65, dep_ratio	9	economy	gdp_pc, inflation	4
settlement, land	urban_pct, forest	9	sectors	agri_va, tax_rev	8
education	sec_enr, educ_exp	5	trade, finance	exports, fdi	5
			transport	two air-traffic counts	2

Full definitions and World Bank codes: `RProject/data/whwide2022.CODEBOOK.md`. 6.52% of the cells are missing and are *not* filled in, which is why `cor()` is called with `use = "pairwise.complete.obs"` above.

3. What we want to know today

The question: today’s notes assert that the largest of many correlations is big *even when nothing is related*, and that the threshold therefore moves as the table gets wider. They assert it, simulate it once at $p = 52$, and never make you derive it. *Where does that number come from, what is it at other widths, and what does it leave standing in a real panel?*

Today: price a scatterplot matrix against a stated budget; measure how far one correlation wanders when there is nothing there; predict the largest of many *before* simulating it; and then spend the answer on the 1326 real correlations.



4. The questions

| Stage 1 — how many panels can you afford?

$\binom{p}{2} = p(p-1)/2$ is the number of *distinct pairs* among p variables — the number of scatterplots in a $p \times p$ grid, and the number of numbers above its diagonal. In R it is `choose(p, 2)`, or `p * (p - 1) / 2`. It is the only part of the grid that grows like p^2 : the diagonal always holds exactly p densities.

Q1. Ten seconds per panel, one 50-minute session, nothing else: what is the largest p whose scatterplot matrix you can afford to look at? guess

Q2. Confirm it by printing the panel count and the minutes for $p = 20, \dots, 30$. R code

Q3. Our panel has $p = 52$. By what factor is it over that budget, and how many sessions would it take? guess

| Stage 2 — what one correlation does when there is nothing there

`rnorm(k)` draws k independent standard normal values and `replicate(B, expr)` evaluates `expr` B times and stacks the results (both are glossed in the notationbox of Lecture 4, §5.1). Two calls to `rnorm(175)` therefore produce two columns that are unrelated *by construction* — not “unrelated as far as we can tell”, but built that way. `set.seed(481)` fixes the stream so the same draws come back tomorrow.

Q4. Two columns of 175 numbers, built to have nothing to do with each other. Their correlation will not be exactly 0. Write down how big you expect $|r|$ to be, typically. guess

Q5. Draw 2000 such pairs, and report the mean and the standard deviation of the 2000 correlations beside $1/\sqrt{n-1}$. R code

Q6. A colleague reports $|r| = 0.20$ between two of our 52 indicators, on all 175 countries, and calls it a weak but real association. Using Q5 only: is he right? guess

| Stage 3 — the largest of many

`matrix(v, nrow = n)` folds a vector into an n -row array and `upper.tri(M)` is TRUE strictly above the diagonal, so `M[upper.tri(M)]` is the vector of the $\binom{p}{2}$ distinct entries — each pair once, no diagonal (Lecture 4, §5.1). `median()` is used rather than the mean throughout this stage because it is the value the largest correlation beats half the time, which is the sentence we want.

Q7. Use the rule to predict, by hand, the largest of the 1326 correlations of a 175×52 table of unrelated columns.

guess

Q8. Now run it: 500 tables of 175×52 unrelated columns, and the median of the 500 largest correlations.

R code

Q9. Do the same at the two widths that matter: $p = 8$ — Lecture 2's table — and the p you found in Q1.

R code

Q10. Two things stop you looking at a wide table: the clock, and the floor. Which one bites first, and at what p ?

guess

Stage 4 — what the floor costs the real panel

`use = "pairwise.complete.obs"` computes each r_{jk} from the countries that report *both* j and k , so different entries of **R** rest on different numbers of countries. The alternative, `use = "complete.obs"`, keeps only countries complete on all 52 — there are 40 of them — and would answer a question about a different world. `setdiff(a, b)` is the elements of **a** that are not in **b**.

Q11. How many of the 1326 real correlations sit inside the floor you just measured, and how many clear it?

R code

Q12. The codebook names four *exact* identities in this panel: `pop0014+pop1564+pop65 = 100`, `rural_pct+urban_pct = 100`, `exports+imports = trade`, and `dep_old+dep_young = dep_ratio`. Remove one column from each and take the five strongest correlations again.

R code

Q13. After that removal, how wide is the panel, how many panels does it have, and what share is still inside the floor?

R code

Q14. Write the one sentence this stage has earned, in the form: “of the 1326 panels, ...”.

guess

Stage 5 — ★ the exact floor — run if the clock allows, first cut otherwise

`qnorm(u)` is the inverse of the standard normal distribution function: the value with area u to its left. It is used here only as arithmetic — to invert a probability we are about to write down — and not as a model for anything.

Q15. Q7's rule overshoot the simulation. Derive the exact value instead: if m independent draws each have standard deviation σ , find the t that the largest absolute value beats exactly half the time.

guess R code

Q16. The exact row and the simulated row still differ, by -0.001 , $+0.004$ and $+0.005$ at the three widths. One of those three is not worth explaining and two are. Say which, and name the assumption behind the two.

guess

5. Conclusion — fill in the blanks

1. A p -variable scatterplot matrix holds _____ panels. At ten seconds each, one 50-minute session pays for _____ of them, so the widest table you can look at in full is $p = \rule{1cm}{0.4pt}$. Ours has $p = 52$, which is _____ sessions.
2. Two unrelated columns of length n have a correlation with mean _____ and standard deviation about _____, which at $n = 175$ is _____ — measured as _____ over 2000 draws.
3. The largest of m such correlations is about $\sigma\sqrt{2\log(2m)}$, which at $m = 1326$ predicts _____; the simulation returns _____, and the exact calculation of Q15 returns _____. The floor at $p = 8$ is _____, and at the width you found in Q1 it is _____, so it _____ with the width of the table.
4. Of the 1326 real correlations, _____ are inside that floor and _____ clear it. Removing the four columns the codebook defines from others leaves the strongest pair at _____, which is _____, not a discovery.
5. So the reason looking fails at $p = 52$ is not that there are too many panels for the _____. It is that the value of “the biggest one I saw” _____, and nothing you can do with your _____ corrects for it.

6. What this opens

Everything on this sheet has been about *pairs*: how many there are, how big the biggest one gets by chance, and how few of ours survive. Every question you would actually ask of these 175 countries — which ones are unlike the rest, and on what — is a question about *rows*, and there is no arrangement of pairwise panels that answers it. Thursday we stop ranking pairs and start ranking countries: **one number per country, computed from all p columns at once**. Note what that immediately buys: one number per country is 175 numbers however large p becomes, and 175 is a list you can read. *Bring this sheet.*